

# SCRIPT ANALISIS NDWI (*NORMALIZE DIFFERENCE WATER INDEX*) MC FEETERS (1996) UNTUK MONITORING BADAN AIR

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Bahasa Pemrograman : Javascript

## NDWI (*Normalized Difference Water Index*)

Normalized Difference Water Index (NDWI) adalah metode yang dikembangkan untuk mendelineasi fitur perairan terbuka dan meningkatkan kehadirannya dalam citra digital penginderaan jauh. NDWI memanfaatkan pantulan radiasi inframerah-dekat (NIR) dan cahaya hijau tampak untuk menonjolkan keberadaan fitur perairan, sekaligus menghilangkan keberadaan tanah dan vegetasi daratan. NDWI juga disarankan dapat memberikan estimasi kekeruhan (turbiditas) badan air menggunakan data digital dari penginderaan jauh (McFeeter, 1996).

## OPRASI NDWI

$$NDWI = (\rho_{Green} - \rho_{NIR}) / (\rho_{Green} + \rho_{NIR})$$

## GOOGLE EARTH ENGINE SCRIPT

```
//-----Cloud Masking
var sentinel2_mask = function(image) {
  var scl = image.select('SCL');
  var mask = scl.neq(3)
    .and(scl.neq(8))
    .and(scl.neq(9))
    .and(scl.neq(10))
    .and(scl.neq(11));
  return image.updateMask(mask);
};

//-----Filter Image
var sentinel2 = ee.ImageCollection('COPERNICUS/S2_SR_HARMONIZED')
  .filterDate('2024-01-01', '2025-01-01')
  .filter(ee.Filter.lt('CLOUDY_PIXEL_PERCENTAGE', 20))
  .map(sentinel2_mask)
  .median()
  .clip(geometry)
  .multiply(0.0001);
Map.centerObject(geometry);

//-----Composite Band
var Composite1182 = sentinel2.select(['B11', 'B8', 'B2']);
var multispectral_param = {min: 0, max: 0.25};
Map.addLayer(Composite1182, multispectral_param, 'Sentinel 2 (11-8-2)');

//-----Visualisasi
var NDWI = sentinel2.normalizedDifference(['B3', 'B8']).rename('NDWI');
var NDWIVis = {
  min: -1, max: 1,
  palette: ['#00441b', '#2a924a', '#78c679', '#c2e699', '#ffffcc',
    '#f7fcfd', '#deebf7', '#9ecae1', '#4292c6', '#084594']
};
Map.addLayer(NDWI, NDWIVis, 'NDWI');
```

```
//-----Klasifikasi
var threshold = 0.009;
var waterMask = NDWI.gt(threshold);
Map.addLayer(waterMask.updateMask(waterMask), {palette: ['blue']}, 'Area NDWI > 0.009');
```

```
//-----Menghitung luas
var pixelArea = ee.Image.pixelArea().divide(10000);
var waterArea = waterMask.multiply(pixelArea);
var totalArea = waterArea.reduceRegion({
  reducer: ee.Reducer.sum(),
  geometry: geometry,
  scale: 10,
  maxPixels: 1e13
});
print('Luas area dengan NDWI > 0.009 (hektar):', totalArea);
```

```
//-----Convert Raster
var waterVector = waterMask.selfMask().reduceToVectors({
  geometry: geometry,
  scale: 10,
  geometryType: 'polygon',
  eightConnected: false,
  labelProperty: 'water',
  maxPixels: 1e13
});
```

```
//-----Menampilkan Layer
Map.addLayer(waterVector, {color: 'cyan'}, 'Vektor NDWI > 0.009');
```

## Referensi

- McFeeters, S.K. The use of the Normalized Difference Water Index (NDWI) in the delineation of open water features. *Int. J. Remote Sens.* 1996, 17, 1425–1432. <https://doi.org/10.1080/01431169608948714>
- Gao, B. NDWI—A normalized difference water index for remote sensing of vegetation liquid water from space. *Remote Sens. Environ.* 1996, 58, 257–266. . [https://doi.org/10.1016/S0034-4257\(96\)00067-3](https://doi.org/10.1016/S0034-4257(96)00067-3)
- Cardille, J. A., Crowley, M. A., Saah, D., & Clinton, N. E. (Eds.). (2024). *Cloud-Based Remote Sensing with Google Earth Engine: Fundamentals and Applications*. Springer International Publishing. <https://doi.org/10.1007/978-3-031-26588-4>